

Andrei Seymour-Howell

Chonnam National University
School of Mathematics
Gwangju, South Korea

andreiseymourhowell@gmail.com
<https://aseymourhowell.github.io/>
<https://github.com/aseymourhowell/>

Employment

2025–Present Postdoctoral Fellow as part of the LFANT group, Chonnam National University.
2023–2025 Research Associate, University of Bristol.

Education

2019–2023 PhD in Mathematics, University of Bristol. Supervised by Andrew Booker.
2015–2019 MMath, University of Nottingham.

Research Interests

Automorphic forms, computational number theory, L-functions, trace formula.

Publications and Preprints

- [1] (With Andrew Booker, Min Lee, David Lowry-Duda and Nina Zubrilina). Murmurations of Maass forms. *Preprint arXiv:2409.00765* 2024
- [2] (With Ce Bian, Andrew Booker, Austin Docherty and Michael Jacobson, Jr.). Unconditional computation of the class groups of real quadratic fields. In *LuCaNT: LMFDB, computation, and number theory*, volume 796 of *Contemp. Math.* Amer. Math. Soc., [Providence], RI, 2024
- [3] Rigorous computation of Maass cusp forms of squarefree level. *Research in Number Theory, ANTS XV collection*, 8, 10 2022

Talks

- June 2026 *Unconditional computation of the class groups of real quadratic fields* - KMS 80th Anniversary Conference, Seoul.
- Apr. 2026 *Numerical computations of Maass cusp forms via the trace formula* - POSTECH Number Theory Seminar.
- Mar. 2026 *Numerical computations of Maass cusp forms via the trace formula* - Kanazawa University Algebra Seminar.
- Oct. 2025 *Numerical computations of Maass cusp forms via the trace formula* - CNUNT Seminar, Chonnam National University.
- Jan. 2025 *Unconditional computation of the class groups of real quadratic fields* - The Great Western Number Theory Seminar (GWNT), University of Exeter.
- Oct. 2024 *Murmurations of Maass Forms* - RIMS Workshop 2024 Analytic Number Theory and Related Topics, Kyoto University.
- Oct. 2024 *Murmurations in Number Theory* - Colloquium Talk, Sophia University, Tokyo.
- Sep. 2024 *Unconditional computation of the class groups of real quadratic fields* - Building Bridges 6, CIRM, Luminy.

- Nov. 2023 *Unconditional computation of the class groups of real quadratic fields* - Linfoot Seminar, University of Bristol.
- July 2023 *Unconditional computation of the class groups of real quadratic fields* - LuCaNT, ICERM, Brown University.
- June 2023 *Rigorous computation of Maass cusp forms of squarefree level* - ArStAFANT Workshop, EPFL.
- Oct. 2022 *Numerical computations of Maass cusp forms* - Canberra Number Theory Seminar, UNSW Canberra.
- Sep. 2022 *Numerical verification of Maass cusp forms* - Automorphic Forms Conference, Erdős Center, Rényi Institute
- Aug. 2022 *Numerical verification of Maass cusp forms* - ELAZ conference, Poznan University.
- Aug. 2022 *Numerical verification of Maass cusp forms* - Algorithmic Number Theory Symposium (ANTS) XV, University of Bristol.
- June 2022 *Numerical verification of Maass cusp forms* - Early Career Researchers in Mathematics, University College London.
- June 2021 *Numerical computations of Maass cusp forms* - Young Researchers in Mathematics, University of Bristol.
- May 2021 *Numerical computations of Maass cusp forms* - London Junior Number Theory Seminar, University College London.
- Apr. 2021 *Numerical verification of Maass cusp forms* - Linfoot Seminar, University of Bristol.
- Mar. 2020 *Numerical computations of the Riemann zeta function* - Postgraduate Seminar, University of Bristol.

Teaching

- 2025 Covered lectures for graduate level course “Analytic number theory”, University of Bristol.
- 2022 Teaching assistant, University of Bristol; 3rd year undergraduate course “Number Theory”.
- 2020 Demonstrator, University of Bristol; 3rd year undergraduate course “Perspectives in Mathematics”.
- 2020 Teaching assistant, University of Bristol; 1st year undergraduate course “Mathematics 1A20”.
- 2019–2020 Teaching assistant, University of Bristol; 1st year undergraduate course “Linear Algebra”.
- 2016–2017 Pass Leader, University of Nottingham.

Conferences attended

- June 2026 *KMS 80th Anniversary Conference* - Seoul.
- May 2026 *Arithmetic and Analytic Properties of Automorphic Forms* - Jeju Island.
- Apr. 2026 *Certain explicit trace formulae on $GL(3)$ intensive lecture series* - Chonnam National University.
- Mar. 2026 *Japan-India Workshop on L-functions and multiple zeta functions* - Sophia University, Tokyo.
- Nov. 2025 *Circle method intensive lecture series* - Incheon National University.
- Oct. 2025 *RIMS Workshop 2025 Analytic Number Theory and Related Topics* - Kyoto University
- Aug. 2025 *Aarhus Automorphic Forms Summer School and Conference* - Aarhus University.
- July 2025 *LuCaNT Conference* - ICERM, Brown University.
- June 2025 *Explicit methods in automorphic forms* - UCL, University College London.
- Jan. 2025 *The Great Western Number Theory Seminar (GWNT)* - University of Exeter
- Nov. 2024 *Murmurations in Arithmetic Geometry and Related Topics* - Simons Center for Geometry and Physics, Stonybrook University
- Oct. 2024 *RIMS Workshop 2024 Analytic Number Theory and Related Topics* - Kyoto University

Sep. 2024 *Building Bridges: 6th EU/US Summer School and Workshop on Automorphic Forms and Related Topics* - CIRM , Luminy.

Aug. 2024 *Automorphic Forms Conference* - Erdős Center, Rényi Institute, Budapest.

July 2024 *Algorithmic Number Theory Symposium (ANTS) XVI* - MIT, Boston.

July 2023 *LuCaNT Conference* - ICERM, Brown University.

July 2023 *Murmurations in Arithmetic Workshop* - ICERM, Brown University.

June 2023 *ArStAFANT Workshop* - EPFL, Lausanne.

May 2023 *Spring school in Arithmetic Statistics* - CIRM, Luminy.

Nov. 2022 *The Great Western Number Theory Seminar* - University of Bristol.

Sep. 2022 *Automorphic Forms Conference* - Erdős Center, Rényi Institute, Budapest.

Aug. 2022 *Automorphic Forms Summer School* - Erdős Center, Rényi Institute, Budapest.

Aug. 2022 *Elementary and Analytic Number Theory (ELAZ) 2022* - Poznan University.

Aug. 2022 *Algorithmic Number Theory Symposium (ANTS) XV* - University of Bristol.

Jul. 2022 *Number Theory Informed by Computation, PCMI Graduate Summer School* - Park City Mathematics Institute, Utah.

Jun. 2022 *Early Career Researchers in Mathematics* - University College London.

Apr. 2022 *The Great Western Number Theory Seminar* - University of Bath.

Apr. 2022 *Warwick-Oxbridge-Manchester-Bristol-London (WOMBL)* - University of Oxford.

Jun. 2021 *Young Researchers in Algebraic and Analytic Number Theory* - University of Bristol.

Jun. 2021 *Young Researchers in Mathematics* - University of Bristol.

Jul. 2020 *Algorithmic Number Theory Symposium (ANTS) XIV* - University of Auckland.

Jun. 2020 *Automorphic Forms Conference* - Erdős Center, Rényi Institute, Budapest.

Nov. 2019 *Young Researchers in Algebraic and Analytic Number Theory* - University of Warwick.

Awards

2018 *IMA prize* - University of Nottingham.

2015–2017 *School achievement prize 3 times* - University of Nottingham.

Service

2025–Present *Co-organiser of Trace Formula learning seminar* - Chonnam National University.

2026 *Co-organiser of Certain explicit trace formulae on $GL(3)$ intensive lecture series by Masao Tsuzuki* - Chonnam National University.

2024–2025 *Co-organiser of Heilbronn Seminar* - University of Bristol.

2022–2023 *Co-organiser of Linfoot Seminar* - University of Bristol.

2021–2022 *Co-organiser of Pre-Heilbronn Seminar* - University of Bristol.

2021 *Co-organiser of Young Researchers in Algebraic and Analytic Number Theory conference 2021* - University of Bristol.

2020 *Co-organiser of MINGLE 2020 (Annual Bristol postgraduate conference)* - University of Bristol.

Programming

Proficient in C, Python, SageMath, Pari/GP and Julia.